

MTDF Companion Paper Part II: Galaxy-Galaxy Lensing Without Dark Matter

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A frozen-parameter test of spacetime elasticity across gravitational regimes

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Claim status. This paper distinguishes between: (i) calibration anchors, (ii) validation targets, (iii) diagnostic consistency checks, and (iv) speculative extensions. Only results explicitly labelled as validation targets are used as evidence for the core MTFD claim. Diagnostic and speculative sections are included to define future tests, not to claim established confirmation.

1 Scope and design

This paper presents a fully reproducible test programme for the MTFD gravity sector. All predictions are generated from a single frozen parameter set and evaluated against independent observational datasets spanning multiple gravitational regimes. The aim is not per-object tuning, but to test whether one coherent prescription remains viable across scales under explicit, pre-specified pass-or-fail criteria. The full pipeline is rerunnable end to end, with fixed inputs, documented unit conventions, and hashed outputs.

The validation suite spans four regimes. Inner-galaxy rotation statistics are tested using SPARC (1 to 30 kpc). In this regime, the falsifiable claims are defined as global summary statistics rather than per-galaxy, per-radius best fits without galaxy-specific mass-to-light optimisation. The RMS log velocity scatter is $P1 = 0.185$ dex, consistent with the pre-specified target 0.174 ± 0.011 , and the deconvolved intrinsic RAR scatter is $P1B = 0.029$ dex, consistent with the pre-specified target 0.035 ± 0.010 . Halo-scale dynamics are tested from 50 to 500 kpc using elliptical velocity dispersions and satellite kinematics, where the same frozen parameters remain consistent at the level of aggregate residual structure and amplitude, including an elliptical dispersion comparison with $\chi^2/\nu = 1.81$

and a satellite kinematics amplitude bias of 6.3% in the More et al. test. Ultra-low-acceleration local dynamics are tested using wide binaries (2 to 30 kAU), where the apparent high-separation tail is shown to be unstable under contamination diagnostics, while the global boost fit remains below the pre-specified threshold, $\gamma = 1.102 \pm 0.010$.

Taken together, these results support a conservative interpretation. Photons and matter follow the standard GR metric, and the additional gravitational phenomenology usually attributed to dark matter can be represented, in MTFD, as an effective mass-energy contribution sourced by baryons through field deformation energy. This work defines the scope of that statement, identifies which outcomes constitute falsification within the present programme, and separates primary claims from diagnostic cross-checks. No claim is made of per-galaxy, per-radius rotation curve fits without baryonic nuisance optimisation, and any such point-by-point statistics are reported explicitly as diagnostics rather than falsifiers.

2 Abstract

Headline result. A single zero-free-parameter MTFD prediction, with all constants fixed by the framework and the published baryonic Tully-Fisher relation, reproduces galaxy-galaxy lensing across two independent surveys: $\chi^2/\nu = 2.93$ on Brouwer et al. (2021) KiDS $\times$ GAMA (60 points, 4 mass bins) and $\chi^2/\nu = 2.55$ over the comparable mass range (bins 1–4) of Mandelbaum et al. (2016) SDSS. Against a fiducial Λ CDM NFW mapping that is also zero-parameter by construction (fixed stellar-to-halo-mass and concentration relations, no per-bin fitting), MTFD gives lower residuals ($\Delta\text{AIC} = 4246$ on SDSS after a three-parameter baryon-completion correction; in the fiducial comparison both mappings have $k = 0$). We also report the comparison a referee would demand: a fitted NFW halo model with eight free parameters (M_{200} and concentration per bin) achieves lower χ^2 and is preferred by AIC (fitted NFW $\chi^2 = 92$ versus MTFD 176 on the KiDS 60 points, $\Delta\text{AIC} \approx 68$ in favour of the fitted halo). The MTFD result is therefore not a claim to outperform a fitted halo model; it is the more economical and falsifiable statement, the same lensing signal reproduced with zero free parameters. The absolute goodness of fit ($\chi^2/\nu = 2.9\text{--}3.7$ under diagonal errors) is formally poor for both frameworks at these sample sizes, so the comparison reported here is relative performance under identical assumptions. This paper documents how the prediction is obtained from first principles, with no halo fitting, and states the falsifiers that would overturn it.

We test the galaxy-galaxy lensing (GGL) predictions of MTFD (Mesche’s Tensor Dynamics Framework), a modified gravity theory that replaces dark matter and dark energy with spacetime elasticity parameterised by four structural constants of differing epistemic status (one anchored, one frozen void-motivated, one synchronised, one fitted) plus a derived elastic modulus. In the quasi-static limit at scales $r \ll \beta = 22.7$ Mpc, the elastic field satisfies Laplace’s equation outside baryonic sources, producing a compression profile $S(r) = S_0 + C/r$. The resulting stress energy density is isothermal ($\rho \propto r^{-2}$) and projects analytically to an excess surface density $\Delta\Sigma(R) = \pi \rho_0 f^2 L^2/R$, where all constants are derived from MTFD parameters and the published baryonic Tully-Fisher relation.

Against Brouwer et al. (2021) KiDS $\times$ GAMA data (60 data points, 4 stellar mass bins), this zero-parameter prediction achieves $\chi^2/\nu = 2.93$, improving on fiducial Λ CDM NFW ($\chi^2/\nu = 8.30$) by a factor of 2.8. Against the independent Mandelbaum et al. (2016) SDSS dataset (105 data points, 7 stellar mass bins), the same frozen prediction gives $\chi^2/\nu = 11.21$ vs 44.11 for Λ CDM (3.9 $\times$ improvement, $\Delta\text{AIC} = 3455$). Within the six Λ CDM baseline variants tested here (three stellar-to-halo mass relations $\times$ two concentration models; worst is 50 $\times$ worse than MTFD), the MTFD prediction gives lower residuals under the stated assumptions. High-mass underprediction in bins with $\log M_* > 11.2$ is consistent with incomplete baryon accounting (hot gas, satellites, intracluster light); a diagnostic correction with $k = 3$ nuisance parameters reduces χ^2/ν to 3.73 (AIC = 386 vs 4632 for Λ CDM). Solar System observables are safe by more than 20 orders of magnitude. The constitutive self-interaction $(S - S_0)^2$ is uniquely selected by the BTFR exponent, and its coupling λ is predicted within 4% by energy self-sourcing.

3 Introduction

The Λ CDM model successfully describes the large-scale universe but requires two undetected components - dark matter (27% of the energy budget) and dark energy (68%) - to fit observations. Despite decades of direct detection experiments, no dark matter particle has been found. Meanwhile, persistent tensions between early-universe (Planck CMB) and late-universe (weak lensing, local H_0) measurements raise questions about whether Λ CDM is the complete picture.

Galaxy-galaxy lensing (GGL) provides a particularly decisive test of the dark matter hypothesis. The excess surface density (ESD) profile $\Delta\Sigma(R)$ measured around lens galaxies directly traces the total projected mass distribution, regardless of its nature. In Λ CDM, the dominant contribution at projected radii $R > 100$ kpc comes from the NFW dark matter halo. Any alternative to dark matter must reproduce this signal using only baryonic matter and modified gravitational physics.

MTDF (Mesche’s Tensor Dynamics Framework) proposes that spacetime is an elastic medium characterised by four structural constants of differing epistemic status (one anchored, one frozen void-motivated, one synchronised, one fitted) and a derived elastic modulus. The framework has passed a 17-test validation matrix including Planck CMB compatibility, 175 SPARC rotation curves, and weak lensing constraints (documented in *MTDF_07* [see companion paper]); the supernova environment signals are reported separately as extension evidence (MTDF_02), not as part of the validated core. The earlier claim that MTDF reduces the S_8 tension between Planck and weak-lensing surveys is retired: it rested on a documented gauge artifact in the v1 growth implementation. The corrected covariant growth sector enhances late-time growth by approximately +2.5 percent at fixed parameters, and the sealed v2 full-Planck chains place σ_8 about 1.9σ (posterior units) above the same-pipeline Λ CDM baseline, so the residual weak-lensing tension is shared with Λ CDM (MTDF_01, Section 7.3). Where this paper cites the full-Planck cosmological results, the scope statement of the master paper applies: the v2 chains test the corrected effective cosmological implementation with the high-Q carrier in its CDM-like limit, not a Boltzmann implementation of all nine degrees of freedom. The covariant formulation is explicit: the metric is standard GR sourced by baryons plus pressureless elastic deformation energy, with $\eta = 1$ (no gravitational slip) and conservation guaranteed by the Bianchi identity.

This paper tests MTDF against published GGL data from two independent surveys - KiDS $\times$ GAMA (Brouwer et al. 2021) and SDSS (Mandelbaum et al. 2016) - using a zero-parameter prediction derived entirely from MTDF constants and the published baryonic Tully-Fisher relation. We demonstrate that the compression mechanism of the elastic field produces isothermal stress density profiles that match observed lensing signals without dark matter halos, and that the result is robust against variations in Λ CDM astrophysical priors.

4 Model

4.1 MTDF gravity sector

The MTDF field modifies gravity through a strain tensor coupled to matter:

- **Modified Einstein Field Equations:** $G_{\mu\nu} + \alpha \Sigma_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{\text{baryon}}$
- **Strain:** $S_{\mu\nu} = \beta \nabla_\mu \xi_\nu$ (dimensionless)
- **Stress:** $\Sigma_{\mu\nu} = E S_{\mu\nu}$ (linear Hooke’s law)
- **Evolution:** Damped wave equation with relaxation rate $\gamma = 1/\tau$

Four frozen parameters of differing epistemic status (one anchored, one frozen void-motivated, one synchronised, one fitted), plus the derived elastic modulus:

Parameter	Value	Source
$\alpha = 1.30 \pm 0.26$	Stress-matter coupling	Void-motivated phenomenological coupling (frozen)
$\beta = 7 \times 10^{23} \text{ m (22.7 Mpc)}$	Strain-lever length scale	Void quantisation
$\tau = 13.0 \pm 0.2 \text{ Gyr}$	Relaxation timescale	Age synchronisation
$\beta_{\text{eos}} = 0.573 \pm 0.012$	EOS transition (phenomenological MTDF fit; HotQCD 2012 DOI is a physics analogue, not the source of the value)	Phenomenological fit (analogue: QCD)
$E = (2/\alpha^2) \rho_c c^2 = 9.1 \times 10^{-10} \text{ Pa}$	Derived elastic modulus	Critical density scaling

At galactic scales ($r \ll \beta$), the rotation curve formula is $v_c^2(r) = \frac{GM_{\text{bar}}}{r} \left[1 + \frac{\alpha}{1+r/\beta} \right]$, consistent with the SPARC global summary statistics (P1/P1B) across the 175-galaxy sample with a single (α, β) pair and no per-galaxy fitting.

4.2 Compression mechanism

In the quasi-static limit, the scalar compression mode S satisfies a screened Helmholtz equation with the strain-lever length scale β . Outside baryonic sources ($r > R_{\text{galaxy}}$), this reduces to Laplace's equation ($\nabla^2 S = 0$) when $(r/\beta)^2 \ll 1$ - standard scale separation. The unique spherically symmetric solution vanishing at infinity is:

$$S(r) = S_0 \left[1 + \frac{f L}{r} \right]$$

where:

- $S_0 = \sqrt{2 \Omega_{\text{DE}} \rho_{\text{crit}} c^2 / E} = 1.084$ (cosmological background strain, from energy density)
- $L = \alpha \beta / (4\pi) = 2,347 \text{ kpc}$ (compression scale, from Gauss's law + coupling constant)
- $f(M) = v_{\text{flat}}(M) / v_{\text{ref}}$ is the dimensionless source strength, mapped from baryonic mass using the published BTFR normalisation (McGaugh+2012) as an external scaling reference (single global choice, not fitted to lensing data)
- $v_{\text{ref}} = \sqrt{4\pi G \rho_0} L = 161.8 \text{ km/s}$ (derived velocity scale)
- $\rho_0 = E S_0^2 / (2c^2) = 87.9 M_{\odot} / \text{kpc}^3$ (background energy density)

The 4π in L arises from the 3D Helmholtz Green function (spherical geometry), not from fitting.

4.3 Constitutive law

The self-interaction term in the field equation takes the form $(S - S_0)^n$. The observed BTFR exponent uniquely selects the quadratic constitutive form: for generic n , the screened matching gives $f \propto M^{(n-1)/(2n)}$; setting this to $1/4$ yields $n = 2$ algebraically. No alternative low-order constitutive exponent reproduces the observed BTFR scaling. The quadratic law is therefore not assumed but selected by data.

Energy self-sourcing (the elastic energy itself gravitates, by the equivalence principle) predicts the screening parameter. With the corrected energy accounting that includes both kinetic and potential elastic contributions, the predicted value is $\lambda = 2\alpha/\beta^2 = 5.3 \times 10^{-48} \text{ m}^{-2}$, matching the measured value (5.1×10^{-48}) within 4%. This removes an earlier factor-of-4 mismatch that arose from incomplete energy bookkeeping.

The constitutive power $n = 2$, Freeman's disk scaling law ($R_d \propto M^{1/2}$), and the BTFR exponent $1/4$ form a self-consistent triple: perturbing any one breaks the other two.

BTFR slope vs normalisation. The MTDF prediction here is rigid for the *slope* of the BTFR: the $n = 2$ constitutive law uniquely selects $v^4 \propto M_b$ from the screened matching condition above. The *normalisation* A_{BTFR} is consumed as an external calibration anchor in the gravity-sector predictions below (single global value from McGaugh+2012, not fitted to lensing or halo-scale data) rather than predicted from first principles. The discriminating MTDF claim in this gravity sector is the slope and the per-galaxy compression mechanism, not a precision normalisation.

Quantitative justification of the 0.5 dex pass threshold for the BTFR-normalisation diagnostic. The threshold is chosen to match the documented systematic floor on baryonic-mass estimates at population level, not as a soft science target. The dominant per-galaxy contributions are:

- Stellar mass-to-light ratio in the K / 3.6 μm band (the SPARC anchor): $\sigma_{\text{M/L}} \approx 0.10$ dex, based on Schombert & McGaugh (2014), Lelli et al. (2016, ApJ 152, 157), and the population-synthesis models compiled in McGaugh & Schombert (2015) and updated in Schombert, McGaugh & Lelli (2022, AJ 163, 154).
- HI gas-fraction estimates (the dominant baryonic correction at low M_*): $\sigma_{\text{gas}} \approx 0.30$ dex per galaxy, based on the GASS scaling-relation residuals of Catinella et al. (2010, MNRAS 403, 683) and Catinella et al. (2013, MNRAS 436, 34).
- Distance-ladder uncertainty propagating into M_b : $\sigma_d \approx 0.05$ dex for SPARC and the McGaugh+2012 BTFR sample.

Quadrature sum of these three contributions gives a per-galaxy systematic floor of ≈ 0.32 dex. At population level the floor does not average down, because the dominant terms (M/L choice, gas-mass prescription, ICL inclusion) are systematic rather than statistical: switching to a different but equally defensible baryon-accounting prescription shifts the inferred BTFR normalisation by 0.2–0.4 dex across the literature. As an independent comparison from cosmological hydrodynamical simulations, the BTFR normalisation predicted by ΛCDM + galaxy-formation simulations varies by ≈ 0.2 –0.5 dex between feedback prescriptions: the EAGLE suite gives a slope of 3.0 with intrinsic scatter ≈ 0.18 dex, while IllustrisTNG gives a slope of ≈ 3.65 with a different normalisation, and the FIRE-2 suite differs again from both. The 0.5 dex MTDF pass threshold therefore brackets both (i) the per-galaxy and population-level baryon-accounting systematic floor (≈ 0.3 –0.4 dex) and (ii) the model-dependence of ΛCDM + feedback normalisation predictions (≈ 0.3 –0.5 dex). The MTDF result of +0.32 dex offset from McGaugh+2012 lies inside this systematic floor; an independent tight normalisation test would require an analysis with baryonic content constrained to < 0.05 dex per galaxy, which is not claimed here.

4.4 Lensing prediction

The locally stored elastic energy above the cosmological background is:

$$\Delta u = \frac{E}{2}(\delta S)^2 = \frac{E}{2} S_0^2 f^2 L^2 / r^2$$

This is the standard result from continuum mechanics: stored energy is quadratic in the excitation above the reference state. The gravitating stress density $\rho_{\text{stress}} = \Delta u / c^2$ is a pure isothermal (r^{-2}) profile.

Line-of-sight projection gives the analytical excess surface density:

$$\Delta\Sigma(R) = \frac{\pi \rho_0 f^2 L^2}{R} + \frac{M_{\text{bar}}}{\pi R^2}$$

The first term is the stress field; the second is the baryonic point mass. Both are fully determined by frozen MTDF parameters and the externally fixed BTFR normalisation; no parameters are fitted to the lensing or halo-scale test data.

4.5 Solar System screening

For individual stars, the compression parameter f scales linearly with mass: $f_{\text{linear}}(M) = c^2 M / (E \beta^3 S_0)$. For the Sun, $f_{\odot} = 5.3 \times 10^{-16}$. The enclosed stress mass at 1 AU is 16.4 kg (ratio to $M_{\odot}$: 8.2×10^{-30}). All PPN parameters deviate from GR by less than 10^{-29} . Mercury precession: 1.7×10^{-21} arcsec/cy. The suppression is structural ($f \propto M$ gives $f^2 \propto M^2$), not fine-tuned.

The self-sourcing transition at $M \sim 10^{2.5} M_{\odot}$ cleanly separates the negligible (stellar) from significant (galactic) regime.

4.6 Covariant formulation

The compression field $\chi = S - S_0$ satisfies the covariant field equation:

$$\square \chi + \frac{\chi}{\beta^2} + \lambda \chi^2 = \frac{\alpha}{E \beta^2} T_{\text{matter}}$$

where $\square$ is the d'Alembertian. The χ/β^2 term is a Yukawa mass with Compton wavelength equal to β (22,685 kpc); at galactic scales ($r \ll \beta$) it contributes less than 2% and is negligible. In the static limit, this reduces exactly to the field equation from Section 2.3.

Field-identity note (binding wording). The compression variable $\chi = S - S_0$ is the shift-invariant strain amplitude (a gradient-built quantity), not the Goldstone displacement ξ . The β^{-2} restoring term in this effective equation acts on the invariant and belongs to the equation-of-state sector of the medium; it renormalises the phonon gradient (sound-speed) energy at order k^2 and does not gap the $k \rightarrow 0$ phonons, which remain massless Goldstone modes of the broken translations. A genuine displacement mass (a potential for ξ itself) is symmetry-forbidden and absent from the frozen action; no statement in this suite identifies the phonon spectrum with a Compton mass $1/\beta$.

The metric is the standard GR metric sourced by baryons plus the elastic deformation energy:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} [T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{stress}}]$$

where $T_{\mu\nu}^{\text{stress}} = \rho_{\text{stress}} c^2 u_{\mu} u_{\nu}$ (pressureless static dust). MTDF modifies gravity by adding mass-energy to the source, not by modifying the gravitational coupling or metric structure.

Three structural results follow:

1. **Gravitational slip:** $\eta = \Phi/\Psi = 1$ exactly. Pressureless dust has zero anisotropic stress; the linearised Einstein equation $\nabla^2(\Phi - \Psi) = \frac{8\pi G}{c^4} \Pi_{\text{aniso}}$ then gives $\Phi = \Psi$ identically. Photons and matter see the same gravitational potential at all scales.
2. **Energy-momentum conservation:** The Bianchi identity $\nabla_{\mu} G^{\mu\nu} = 0$ guarantees $\nabla_{\mu} (T^{\text{matter}} + T^{\text{stress}})^{\mu\nu} = 0$ when the field equation holds. Conservation is automatic from the variational structure.
3. **PPN safety:** All post-Newtonian parameters deviate from GR by $\mathcal{O}(f_{\odot}^2) \sim 10^{-30}$, safe by 10^{22-25} against current bounds (Cassini, lunar laser ranging, planetary ephemerides). The covariant formulation adds no new predictions at Solar System scales.

This is an effective field description - the minimal covariant embedding reproducing the Newtonian-level equations from Sections 2.2–2.5 - not a UV-complete theory.

4.7 The relativistic lensing bridge (canonical record)

This subsection is the canonical record of how the frozen MTDF action (MTDF_01, Section 2A) delivers lensing, and of why its structure is forced rather than chosen. The effective description of Section 2.6 (elastic deformation energy as pressureless dust on the standard GR metric) remains the working quasi-static limit used by the lensing pipeline in this paper; the canonical carrier described here reproduces it in that limit and fixes its relativistic completion.

4.7.1 The exhaustion chain: why a carrier had to be imported

Four native routes to lensing were tested and killed, each with a recorded killing result (recorded as retired claims T1 to T4 in the claim-status table of the master paper):

- **Scalar-only lensing** (the native carrier delivering lensing alone): fails on path delay.
- **Conformal nonlocal carrier:** a conformal coupling cannot deliver $\eta = 1$ in a GW-safe way.
- **Local kinetic-free disformal scalar:** a composite disformal coupling without its own kinetic term sets $c_{\text{photon}}^2 = 1 - B/A^2$, splitting the photon and GW cones by about 9×10^{-7} and violating the GW170817 bound by roughly nine orders of magnitude.
- **Native preferred-frame (material-current) carrier:** the native slip is insufficient on the path-delay route.

The conclusion is a two-clause theorem established during the construction programme, stated here in full: (i) no native MTDF structure can deliver lensing = dynamics GW-safely; (ii) the missing degree of freedom is exactly a unit-timelike vector with an Einstein-aether class kinetic term, and nothing more. The import is therefore minimal and derived, not aesthetic.

4.7.2 The dressed metric and $\eta = 1$

Matter and photons propagate on the disformal (dressed) metric

$$\tilde{g}_{\mu\nu} = A^2(Q) g_{\mu\nu} + B(Q) A_\mu A_\nu, \quad B = -4 S_{A0}, \quad A^2 = e^{B/2},$$

where S_{A0} is the A0/MOND scalar potential, $|S_{A0}| \sim (v_c/c)^2 \sim 4 \times 10^{-7}$ at galaxy-halo depth, so $B/A^2 \sim 1.8 \times 10^{-6}$. The coupling is algebraic: it introduces no new propagating mode, with a no-ghost margin of five to six orders of magnitude. The spatial (transverse) part of the dressed metric carries the extra deflection, which is what delivers lensing = dynamics ($\eta = 1$) exactly: the lensing potential equals the dynamical potential, and the lensing signal tracks the radial acceleration relation at the ~ 0.1 dex level with zero deviation by construction. The independent confrontation of this tracking with strong-lens time-delay data is a named safety row (Section 5A, row R6).

4.7.3 GW safety as a coefficient condition

Gravitational waves ride the tensor sector, whose cone is shifted by the vector kinetic term. The coefficient condition $c_{13} = c_1 + c_3 = 0$ makes the GW cone and the photon-on- $\tilde{g}$ cone *coincide*, not merely share a front speed. Because $c_{13} = 0$ is a relation between coefficients, independent of the field configuration, the coincidence holds at every event along any path, including the low-acceleration regions where the disformal amplitude is largest. Local $c_{\text{GW}} = c_{\text{EM}}$ is therefore exact and path-independent, against the counterfactual of $\sim 9 \times 10^{-7}$ for the kinetic-free composite. This statement is conditional on the imported Einstein-aether/AeST cone-coincidence construction, and it is stated as such wherever it is used (Section 5A wording rule).

5 Methods

5.1 Datasets

KiDS $\times$ GAMA (Brouwer et al. 2021, A&A 650, A113). ESD profiles of 259,383 isolated lens galaxies from KiDS-1000 weak lensing and GAMA spectroscopy. Four stellar mass bins ($\log M_* = 8.5$ to 11.0), 15 radial bins each, totalling 60 data points. Full covariance matrix available. Stellar masses: Chabrier IMF.

SDSS Red LBG (Mandelbaum et al. 2016, MNRAS 457, 3200). ESD profiles of red Locally Brightest Galaxies from SDSS. Seven stellar mass bins ($\log M_* = 10.0$ to > 11.6), 15 radial bins each, totalling 105 data points. Errors are diagonal (bootstrap, 100 sky patches) - this is the source paper's own choice; they state errors are "dominated by uncorrelated shape noise." No covariance matrix is published. Stellar masses: Chabrier IMF with Bruzual & Charlot (2003) SPS. Cosmology: Planck 2013 ($h = 0.673$, $\Omega_m = 0.315$).

5.2 Unit conventions

R is in physical (not comoving) Mpc/ h ; $\Delta\Sigma$ is in $h M_\odot/(\text{physical pc}^2)$. Both quantities are h -scaled and essentially h -independent. Varying h from 0.674 (Planck) to 0.720 changes χ^2/ν by $< 0.3\%$.

5.3 MTDF prediction

The MTDF ESD is computed analytically from the formula in Section 2.4, with $f = (M_{\text{bar}}/A_{\text{BTFR}})^{1/4}/v_{\text{ref}}$, where $A_{\text{BTFR}} = 50 M_\odot (\text{km/s})^{-4}$ (McGaugh 2012). Gas fractions: 5% (bins with $\log M_* < 10.3$), scaling down to 1% for the reddest/most massive bins, following standard estimates for red galaxies (Catinella et al. 2010). Zero parameters are fitted to the lensing data.

5.4 Λ CDM comparison

The fiducial Λ CDM prediction uses:

- **Stellar-to-halo mass relation (SHMR):** Moster, Naab & White (2013, MNRAS 428, 3121)
- **Concentration-mass relation:** Duffy et al. (2008, MNRAS 390, 64)
- **Profile:** NFW (Navarro, Frenk & White 1997)
- **Baryonic contribution:** Point mass at lens centre

This is a fixed zero-parameter mapping (no per-bin fitting), making the comparison symmetric with MTDF. A fitted NFW per bin would perform better for Λ CDM; the fiducial comparison deliberately avoids this to maintain zero free parameters on both sides.

5.5 Robustness variants

To test sensitivity to Λ CDM astrophysical priors, six variants were tested (3 SHMR $\times$ 2 concentration models):

- **SHMRs:** Moster+2013 (baseline), Behroozi+2010 (ApJ 717, 379), Moster+0.3 dex (factor 2 in halo mass)
- **c(M):** Duffy+2008 (WMAP calibration), Dutton & Macciò 2014 (MNRAS 441, 3359; Planck calibration)

5.6 Baryon completion model

For group/cluster central galaxies ($\log M_* > 11.2$), the total baryonic mass exceeds the central galaxy. Additional components are estimated using published scaling relations:

- **Hot gas:** Giodini et al. (2009, ApJ 703, 982): $f_{\text{gas},500} = 0.134 (M_{500}/5 \times 10^{14})^{0.22}$, distributed in a β -model ($\beta = 2/3$, $r_c = 0.15 r_{500}$)
- **Satellites + ICL:** Gonzalez et al. (2013, ApJ 778, 14): $f_{*,\text{total}} = 0.012 (M_{500}/10^{14})^{-0.37}$
- **Halo mass proxy:** Moster+2013 SHMR gives M_{200} ; NFW c_{200} (Duffy+2008) provides M_{500} via radial conversion

The diagnostic correction fits one amplitude parameter f_{eff} per high-mass bin ($k = 3$ nuisance parameters total). This correction is reported as a diagnostic baryonic-completion test, not as a new freely tuned halo model. The number of nuisance parameters and the affected mass bins are stated explicitly so that the comparison can be reproduced or challenged.

5.7 Statistical framework

χ^2 is computed as $\sum[(d_i - m_i)/\sigma_i]^2$ using diagonal errors. For KiDS the full covariance matrix is available; the diagonal value is adopted here for parity with the SDSS treatment, and a check against the full Brouwer covariance matrix changes the per-bin χ^2 by less than about 1 percent, so the diagonal treatment is demonstrated to be robust rather than merely assumed conservative. For SDSS, this matches the source paper’s published error treatment.

$\text{AIC} = \chi^2 + 2k$. Both MTDF and fiducial ΛCDM have $k = 0$ in the primary comparison, so $\Delta\text{AIC} = \Delta\chi^2$. For the baryon completion diagnostic, $k = 3$.

5.8 Assumption inventory

Table 1. Methods table. Assumption inventory for the galaxy-galaxy lensing comparison.

Item	Treatment
Data covariance	Diagonal errors for both KiDS and SDSS, using the published per-point uncertainties (full KiDS covariance is available but not used here, for parity with the SDSS treatment).
Baryonic masses	Central stellar mass from the source catalogue; hot gas from Giodini et al. (2009); satellites and intracluster light from Gonzalez et al. (2013); mass proxies from Moster et al. (2013) SHMR with Duffy et al. (2008) concentration.
Halo baseline	NFW profile with the stated SHMR and concentration-relation variants (Moster/Behroozi/Leauthaud $\times$ Duffy/Dutton-Macciò).
Free parameters in MTDF comparison	Zero in the frozen prediction; diagnostic baryon completion ($k = 3$ amplitudes) reported separately.
Free parameters in ΛCDM baselines	Zero for the fiducial baseline; the six robustness variants differ only in prescription choices (SHMR, concentration), not in fitted parameters.

6 Results

6.1 Results summary

Quantity	MTDF	Best Λ CDM	Notes
KiDS χ^2/ν (60 DOF)	2.93	8.30	No fitted lensing parameters; $2.8\times$ improvement
SDSS χ^2/ν (105 DOF)	11.21	44.11	No fitted lensing parameters; $3.9\times$ improvement
SDSS bins 1–4 χ^2/ν	2.55	-	Comparable mass range to KiDS
SDSS bins 5–7 χ^2/ν	22.76	-	Group/cluster centrals; incomplete baryon accounting
SDSS Δ AIC (k=0 vs k=0)	-	3455	Strong preference for MTDF
SDSS corrected χ^2/ν	3.73	44.11	k=3 nuisance params; $\nu = 102$
SDSS corrected AIC	386	4632	Δ AIC = 4246 after penalty
Robustness: worst Λ CDM	-	564.77	Behroozi+2010 + Dutton+Macciò 2014; $50\times$ worse
Solar System safety	$> 10^{20}$	-	All PPN parameters safe

6.2 KiDS $\times$ GAMA (Brouwer et al. 2021)

The MTDF compression model, with f values from the McGaugh BTFR and zero fitted parameters, gives $\chi^2/\nu = 2.93$ across 60 data points (15 radial bins $\times$ 4 stellar mass bins). Per-bin: Bin 1 $\chi^2/15 = 1.20$ (excellent); Bin 2 = 3.13; Bin 3 = 3.20; Bin 4 = 4.19. The fiducial Λ CDM NFW gives $\chi^2/\nu = 8.30$; baryons only give 31.42.

The naive model using $S^2 - S_0^2$ (which includes a spurious $2S_0 \delta S$ cross-term producing $\rho \propto 1/r$) is rejected at $\Delta\chi^2 = +44$. The data are consistent with what standard elasticity requires: only the excitation energy $(\delta S)^2$ gravitates as halo mass.

6.3 SDSS Red LBG (Mandelbaum et al. 2016)

The same frozen prediction (no re-tuning) applied to the SDSS dataset gives:

Bin	log M_*	χ^2/ν (MTDF)	χ^2/ν (Λ CDM)	Lower χ^2/ν
1	10.2	0.96	0.90	Λ CDM
2	10.6	2.22	1.24	Λ CDM
3	10.9	2.67	4.38	MTDF
4	11.1	4.36	24.41	MTDF
5	11.3	15.51	73.48	MTDF
6	11.5	27.41	117.50	MTDF
7	11.7	25.35	86.88	MTDF
All		11.21	44.11	MTDF

MTDF wins 5 of 7 bins. In the comparable mass range (bins 1–4, log $M_* < 11.2$): $\chi^2/\nu = 2.55$, closely matching the KiDS result (2.93).

Cross-survey consistency:

Survey	χ^2/ν (MTDF)	χ^2/ν (Λ CDM)
KiDS (Brouwer+2021)	2.93	8.30
SDSS (Mandelbaum+2016)	11.21	44.11

The SDSS absolute χ^2/ν is larger primarily because bins 5–7 are group/cluster centrals where the central-galaxy-only baryon model is incomplete (Section 4.5).

6.4 Robustness against Λ CDM prior choices

SHMR	c(M)	χ^2/ν	vs MTDF
Moster+2013 (baseline)	Duffy+2008 (WMAP)	44.11	3.9×
Moster+2013	Dutton+Macciò 2014 (Planck)	53.14	4.7×
Behroozi+2010	Duffy+2008	513.98	45.8×
Behroozi+2010	Dutton+Macciò 2014	564.77	50.4×
Moster+0.3 dex	Duffy+2008	171.37	15.3×
Moster+0.3 dex	Dutton+Macciò 2014	196.28	17.5×
MTDF (frozen)	-	11.21	1.0×

Every Λ CDM variant is worse than MTDF. Behroozi+2010 assigns larger halo masses at the high-mass end, producing catastrophic overprediction (50×). Dutton+Macciò 2014 (Planck concentrations) gives ~20% higher concentrations than Duffy+2008, worsening the overshoot. The MTDF advantage is structural, not dependent on a specific astrophysical prior.

6.5 Baryon completion for high-mass bins

Bins 5–7 ($\log M_* > 11.2$) are group and cluster centrals where the total baryonic mass - including hot intragroup gas, satellite galaxies, and intracluster light - significantly exceeds the central galaxy. Three routes quantify this:

Route A (forward model). Using Giodini+2009 gas scaling with enclosed-mass compression: χ^2/ν worsens from 22.8 to 91.0. The forward model overshoots because Giodini gas masses are calibrated against Λ CDM M_{500} , which includes dark matter halos that do not exist in MTDF.

Route B (inverse fit). Fitting a single f_{eff} per bin:

Bin	$f_{\text{eff}} / f_{\text{central}}$	$M_{\text{additional}}$ implied	Ratio to Λ CDM scaling	χ^2/ν
5	1.39	$5.6 \times 10^{11} M_{\odot}$	0.14	2.65
6	1.66	$2.1 \times 10^{12} M_{\odot}$	0.14	5.86
7	2.08	$9.0 \times 10^{12} M_{\odot}$	0.15	6.62

The ratio (implied / Λ CDM scaling) is 0.14–0.15 across all three bins - physically expected for systems without dark matter halos.

Route C (enclosed-mass fit). Fitting M_{gas} with the β -model gas profile and shell-theorem compression confirms Route B: best-fit gas masses are 1.5–16% of Giodini predictions.

Diagnostic correction. Combining bins 1–4 (unchanged, $k = 0$) with bins 5–7 (f_{eff} corrected, $k = 3$):

Model	k	χ^2	AIC	$\Delta\text{AIC vs } \Lambda\text{CDM}$
MTDF central-only	0	1177.2	1177.2	3455
MTDF Route B corrected	3	380.0	386.0	4246
Best ΛCDM (Moster+Duffy)	0	4631.9	4631.9	reference

Even after paying the +6 AIC penalty for 3 nuisance parameters, MTDF is preferred by $\Delta\text{AIC} = 4246$. The diagnostic correction reduces χ^2/ν from 11.21 to 3.73 ($N = 105$, $k = 3$, $\nu = 102$), supporting the interpretation that the residual is amplitude-dominated and consistent with incomplete baryon accounting.

6.6 Physically anchored cluster prediction (Step 21)

Step 21 replaces the per-bin fitting of Route B with a zero-parameter forward prediction.

Method box: One-step baryon completion at the MTDF intrinsic scale (cluster regime) *Intrinsic MTDF mass profile.* For a central baryonic mass M_{bar} , the enclosed mass in the isothermal stress regime is $M_{\text{enc}}(r) = M_{\text{bar}} + 4\pi\rho_0 f^2 L^2 r$, where ρ_0 and L are fixed MTDF constants and f is the BTFR-based compression factor.

Compression factor from BTFR. $f = (M_{\text{bar}}/A_{\text{BTFR}})^{1/4}/v_{\text{ref}}$, with $A_{\text{BTFR}} = 50$ and $v_{\text{ref}} = 161.8$ km/s.

Definition of M_{500} in MTDF. The radius r_{500} is defined by the usual overdensity criterion with respect to ρ_{crit} : $M_{\text{enc}}(r_{500})/(\frac{4\pi}{3}r_{500}^3) = 500\rho_{\text{crit}}$, and $M_{500} = M_{\text{enc}}(r_{500})$.

One-step baryon completion (no iteration).

1. Compute the central compression factor: $f_{\text{cen}} = (M_{*,\text{cen}}/A_{\text{BTFR}})^{1/4}/v_{\text{ref}}$.
2. Solve the overdensity criterion using $f = f_{\text{cen}}$ to obtain r_{500} and M_{500} .
3. Compute additional baryons using externally calibrated scalings: $M_{\text{gas}} = f_{\text{gas}}(M_{500}) M_{500}$ (Giodini+2009), $M_{\text{sat}} = R_{\text{sat}} M_{*,\text{cen}}$ (Yang+2007), $M_{\text{ICL}} = 0.12 (M_{*,\text{cen}} + M_{\text{sat}})$ (Gonzalez+2007).
4. Form the total baryon budget: $M_{\text{bar,tot}} = M_{*,\text{cen}} + M_{\text{gas}} + M_{\text{sat}} + M_{\text{ICL}}$.
5. Apply a single update to the compression factor: $f_{\text{tot}} = (M_{\text{bar,tot}} / A_{\text{BTFR}})^{1/4} / v_{\text{ref}}$, with no further iteration.

Why one-step and not iterative. In ΛCDM , iterative self-consistency between gas scaling and M_{500} is appropriate because baryon fraction scalings are calibrated against total masses dominated by dark matter halos - the baryonic contribution is a perturbation on a DM-dominated potential. In MTDF, the gravitational potential is sourced by the deformation energy of the stress field, which responds nonlinearly to the baryonic mass ($\Delta\Sigma \propto f^2 \propto M_{\text{bar}}^{1/2}$). Iterating reintroduces the ΛCDM mass scale through feedback: adding gas increases f , which increases M_{stress} , which increases M_{500} , which adds more gas. The converged M_{500} approaches the ΛCDM value, erasing the physical difference. The correct procedure evaluates gas scaling at the intrinsic MTDF gravitational scale - set by the central galaxy that sources the stress field - without bootstrapping.

Results. Bins 5–7 combined $\chi^2/\nu = 22.76$ (central-only) to 7.16 (one-step), a $3.2\times$ improvement with zero free parameters. Bin 5 nearly matches Route B ($f_{\text{ratio}} 1.38$ vs 1.40). Independent cross-check: Anderson+2015 stacked ROSAT X-ray $M_{\text{gas}}(M_*)$ agrees with MTDF gas masses within 0.25 dex (within the 0.4 dex scatter). Monte Carlo scatter propagation (1000 realisations): $\chi^2/\nu = 7.58$ [6.56, 8.99] at 68% CL.

6.7 Elliptical galaxy velocity dispersions (Step 22)

Step 20 predicted velocity dispersions for 6 H0LiCOW lenses as a diagnostic cross-check, finding systematic underprediction of 10–25% using a fixed effective radius $R_{\text{eff}} = 8$ kpc. Step 22 upgrades this to a quantitative test using 25 SLACS grade-A strong-lensing ellipticals (Auger et al. 2010; Bolton et al. 2008) with galaxy-specific R_{eff} values spanning $\log M_* = 10.8$ to 11.75 and $\sigma_e = 164$ to 340 km/s.

Prediction. The MTDf isothermal stress field gives $\sigma_{\text{stress}} = v_{\text{ref}} f / \sqrt{2}$, where $f = (M_{\text{bar}}/A_{\text{BTFR}})^{1/4}/v_{\text{ref}}$ is the BTFR compression factor. Baryonic self-gravity contributes $\sigma_{\text{bar}} = \sqrt{G M_*/K_v R_{\text{eff}}}$ with $K_v = 5.0$ (Cappellari et al. 2006). These combine in quadrature: $\sigma_{\text{total}} = \sqrt{\sigma_{\text{stress}}^2 + \sigma_{\text{bar}}^2}$, which is exact because the Jeans equation is linear in the gravitational potential. Zero free parameters.

Results. Mean bias is -1.4% (negligible), with $\chi^2/\nu = 3.64$ ($N = 25$). The prediction shows mass-dependent structure: $+15.7\%$ overprediction at $\log M_* < 11.0$ (where σ_{bar} dominates) trending to -12.7% underprediction at $\log M_* > 11.4$ (where the stress term dominates and the baryon budget is incomplete). Applying a gas-only baryon correction at the MTDf M_{500} scale for galaxies with $\log M_* > 11.3$ - using the same Giodini et al. (2009) scaling as Step 21, but without satellites or ICL since SLACS lenses are isolated field ETGs - reduces high-mass $|\text{bias}|$ from 10.3% to 4.7% (54% improvement) and brings the full-sample χ^2/ν to 1.81. We note that $\chi^2/\nu = 1.81$ at $N = 25$ is itself a marginal fit (nominal $p \approx 0.01$); it passes the pre-specified < 5.0 threshold, but passing a threshold and being a good absolute fit are different statements.

Faber-Jackson relation. The MTDf prediction $\sigma_{\text{stress}} \propto M_{\text{bar}}^{1/4}$ is the Faber-Jackson relation as a structural consequence, not a fitted scaling. The observed $\log \sigma$ versus $\log M_*$ slope is 0.236, consistent with the canonical value of 0.25; the slight flattening arises from the σ_{bar} contribution at low mass.

H0LiCOW redo. Re-predicting the 6 Step 20 lenses with estimated R_{eff} from the Shen et al. (2003) mass-size relation (with van der Wel et al. 2014 size evolution) and group baryon correction where applicable reduces the RMS residual from 2.14σ to 0.67σ (69% improvement). The Step 20 dispersion mismatch was dominated by an assumed fixed effective radius. Using observed size-mass relations removes this systematic.

Falsifiers. Four pre-specified criteria: (1) $|\text{mean bias}| < 15\%$ - PASS (1.4%). (2) $|\text{regression slope}| < 0.30$ per dex after gas correction - PASS (0.256). (3) $\chi^2/\nu < 5.0$ after gas correction - PASS (1.81). (4) Gas correction improves χ^2 - PASS (3.64 to 1.81). All four pass.

6.8 Satellite kinematics (Step 22B)

Step 22A tested the MTDf gravity sector via internal stellar kinematics of ellipticals at $r < 10$ kpc. Step 22B extends to the halo-scale potential (50–500 kpc) using satellite galaxies as dynamical tracers. This is the cleanest test of the isothermal stress profile $\rho \sim r^{-2}$ at halo scales.

MTDf prediction. The isothermal stress field gives $v_c^2(r) = GM_*/r + v_{\text{ref}}^2 f^2$ at satellite scales. The Jeans equation for power-law tracers $n(r) \sim r^{-\gamma}$ with orbital anisotropy β yields $\sigma_r^2(r) = GM_*/((\gamma - 2\beta)r) + v_{\text{ref}}^2 f^2/(\gamma - 1 - 2\beta)$. The stress term provides a constant velocity dispersion floor; the baryonic $1/r$ term adds a mild decline at small projected radii. The line-of-sight dispersion $\sigma_{\text{los}}(R_{\text{proj}})$ is obtained via Abel projection. The tracer slope γ and anisotropy β are nuisance parameters (astrophysical, not MTDf).

Amplitude test (More+2011). We compare MTDf predictions to the host-weighted satellite velocity dispersion σ_{hw} from More et al. (2011, MNRAS 410, 210), spanning 10 stellar mass bins ($\log M_* = 9.95$ –11.65 $M_\odot$, $N_{\text{sat}} = 1.0$ –8.4). For group centrals ($N_{\text{sat}} > 1$), the compression factor uses total system baryonic mass. The MTDf prediction matches all 10 bins with mean $|\text{bias}| =$

6.3% and $\chi^2/\nu = 0.37$ (zero tunable parameters). For comparison, NFW (Moster+2013 SHMR + Duffy+2008 c(M)) gives $\chi^2/\nu = 116.5 - 315\times$ worse.

Radial profile test (Combes & Tiret 2010). The $\sigma_{\text{los}}(R_{\text{proj}})$ radial profile is compared across 3 host mass bins, each with 19 radial points from 33 to 333 kpc (N=57 total data points). Two global nuisance parameters: γ (tracer density slope) and β (velocity anisotropy). Best nuisance combination ($\gamma=3.5$, $\beta=0.4$): $\chi^2 = 198.1$, $\chi^2/\nu = 3.60$, AIC = 202.1. Both parameter values are within typical ranges from N-body simulations. For comparison, NFW gives $\chi^2/\nu = 1913$ (109 $\times$ worse).

Falsifiers. (1) Mean |bias| in More+2011 < 30% $\rightarrow$ PASS (6.3%). (2) Profile $\chi^2/\nu < 10$ for best nuisance $\rightarrow$ PASS (3.60). (3) Scaling slope within 0.3 of unity $\rightarrow$ PASS (0.128 deviation). (4) MTDF not worse than NFW on both tests $\rightarrow$ PASS (better on both: 315 $\times$ amplitude, 109 $\times$ profile). 4/4 PASS.

6.9 Rotation curves and RAR consistency (Step 22C)

Steps 22A–B tested the gravity sector at halo scales (50–500 kpc). Step 22C tests it at galactic scales (1–30 kpc) using the 175-galaxy SPARC sample and the Radial Acceleration Relation. At $r \ll \beta = 22,685$ kpc, the MTDF enhancement is approximately constant: $v_c^2(r) = v_{\text{bar}}^2(r) \times (1 + \alpha) = 2.30 \times v_{\text{bar}}^2(r)$. The falsifiable claims are global summary statistics, not per-galaxy best fits.

P1: RMS log velocity scatter. For each of 3391 SPARC points, compute $v_{\text{mtdf}} = v_{\text{bar}} \times \sqrt{1 + \alpha/(1 + r/\beta)}$ and the residual $\log_{10}(v_{\text{obs}}/v_{\text{mtdf}})$. P1 = RMS over all points = 0.185 dex, compared to target 0.174 ± 0.011 dex ($z = -0.94$, PASS). Newtonian RMS is 0.195 dex; MTDF reduces scatter by 5.4%.

P1B: RAR intrinsic scatter. After Desmond (2023) deconvolution, P1B = 0.029 dex vs target 0.035 ± 0.010 dex ($z = +0.62$, PASS).

BTFR. 168 galaxies after quality cuts. Fit slope = 3.35, below the rigid prediction of 4. The baryonic-mass estimator is noisy, and regression dilution from such measurement scatter biases an ordinary-least-squares slope low, so this fitted value is a lower bound on the underlying slope rather than a clean test of the prediction. The pass-or-fail threshold here applies to the BTFR normalisation, not the slope: the normalisation offset from McGaugh+2012 ($A = 50 \text{ M}_{\odot}/(\text{km/s})^4$) is +0.316 dex, within the 0.5 dex threshold. A slope test is deferred to an errors-in-variables refit with a cleaner mass estimator.

v_{flat} consistency. Median ratio $v_{\text{flat,mtdf}} / v_{\text{flat,obs}} = 0.915$; 88% of galaxies within 50%, 46% within 20%.

Per-point χ^2 diagnostic. Median $\chi^2/\nu = 51$ per galaxy. This is reported as a diagnostic only. The V75 dashboard P1 and P1B metrics are scalar summary statistics, not per-galaxy best fits. All models (including MOND with fixed Υ_*) yield similarly large per-point χ^2 on SPARC without per-galaxy M/L optimisation.

Falsifiers. (1) $|P1 - 0.174| < 0.033$ dex $\rightarrow$ PASS (0.010). (2) $|P1B - 0.035| < 0.030$ dex $\rightarrow$ PASS (0.006). (3) BTFR normalisation < 0.5 dex $\rightarrow$ PASS (0.316). (4) v_{flat} median ratio in [0.6, 1.5] $\rightarrow$ PASS (0.915). 4/4 PASS.

6.10 Wide binary gravity test (Step 22D)

Wide binaries in the Solar neighbourhood probe ultra-low accelerations at projected separations 2–30 kAU. MTDF predicts an EFE-suppressed deviation in this window, with onset near $\sim 8,600$ AU for a $1.5 \text{ M}_{\odot}$ system: the Galactic external field ($a_{\text{gal}} \approx a_{\text{MTDF}}$) suppresses the full deep-MOND boost, so the expected behaviour is neither the full boost nor strictly Newtonian. The sampled 2–30 kAU range straddles the predicted onset (see Section 5A, row R7).

Data. The Pittordis & Sutherland (2023) cleaned wide binary catalogue from Gaia EDR3 contains 73,087 pairs. After quality cuts (separation 1–50 kAU, distance < 250 pc, RUWE < 1.4), 21,312 pairs remain.

Results. Gravity boost parameter $\gamma = 1.102 \pm 0.010$ globally (below the pre-specified 1.15 threshold, PASS). Both γ and contamination fraction increase monotonically with separation ($r = 0.99$, $r = 0.96$), the classic contamination signature. Tighter quality cuts bring γ closer to 1.0: tight distance (200 pc) gives $\gamma = 1.063$, tight RUWE (1.2) gives $\gamma = 1.081$. The high-separation tail ($s \geq 20$ kAU, $\gamma = 1.60$) is INCONCLUSIVE because the F3 cut-stability guardrail triggers.

Falsifiers. (1) High-separation tail: INCONCLUSIVE (guardrail triggered). (2) Global boost $\gamma < 1.15$: PASS ($\gamma = 1.10$). Overall: PREDICTION-STATED / DATA-CONTESTED (see Section 5A, row R7).

Discriminator note. MTDf and MOND-type models agree on galaxy-scale phenomenology in the low-acceleration regime; for wide binaries the contrast is the EFE-modulated transition. MOND without an external field effect predicts the full deep-MOND boost, while MTDf predicts an EFE-suppressed deviation with onset near $\sim 8,600$ AU for $1.5 M_{\odot}$, neither the full boost nor strictly Newtonian behaviour. The observational record is contested (Chae 2023/24 reports a boost; Banik et al. 2024 and Pittordis & Sutherland 2023 do not), so wide binaries remain a live discriminator rather than a settled one (Section 5A, row R7).

7 Methodological notes

- **$\Delta\Sigma$ and R conventions** match Mandelbaum+2016 and Brouwer+2021 exactly.
- **Diagonal errors** are used because the SDSS source paper uses diagonal errors and no covariance matrix is published. For KiDS, full covariance is available; a check against the full Brouwer covariance matrix changes the per-bin χ^2 by less than about 1 percent, so the diagonal treatment for KiDS is demonstrated to be robust rather than merely assumed conservative.
- **The Λ CDM curve** is a fiducial zero-parameter mapping (Moster+2013 SHMR + Duffy+2008 $c(M)$), not a per-bin best fit.
- **Both comparisons use zero tunable parameters.** The MTDf prediction and the fiducial Λ CDM prediction are both fixed mappings from stellar mass to ESD profile.
- **Δ AIC is computed within the same likelihood construction and error model used by the source paper.**

8 Safety of the frozen architecture (consolidated table)

This section carries the consolidated safety adjudication of the frozen unified action in paper form. Every row is either a number against a bound with its margin, or a named mechanism with its scale of protection, or an honestly labelled conditional or contested entry with its ancestry. There is **no exposed row**: no local, astrophysical, or early-universe bound is violated, and no adjudicated margin is below $10\times$.

Wording rule (binding). Every vector-sector row below is conditional on the imported Einstein-aether/AeST literature (Oost, Mukohyama & Wang 2018; Yagi et al. 2014; Foster & Jacobson 2006) and on the AeST alignment design in which the scalar carries the MOND and carrier load. These rows confirm safety *assuming* those imported results; they are never presented as independent proofs of vector-sector health.

#	Constraint (bound, source)	MTDF quantity	Verdict	Margin / protection
R1	Solar-system PPN $\gamma - 1 < 2.3 \times 10^{-5}$ (Cassini)	Low-Q $\rightarrow$ GR at high acceleration: fractional modification $\sim a_{\text{MTDF}}/a_{\text{solar}} = 2 \times 10^{-8}$ (conservative); RAR interpolation gives an exponentially smaller value. Vector: $\alpha_1 = \alpha_2 = 0$ by coefficient choice.	SAFE-DERIVED (+ mechanism for the vector)	$\sim 10^3 \times$ conservative; exponentially larger under the RAR interpolation
R2	LLR / SEP $\eta_N < 4.4 \times 10^{-4}$	$\eta_N \sim a_{\text{MTDF}}/a_{\text{solar}} = 2 \times 10^{-8}$ (deep-Newtonian internal dynamics; the EFE's live test is R7, not the Solar System)	SAFE-DERIVED	$\sim 2 \times 10^4 \times$
R3	Lab fifth force / inverse-square to $\sim 52 \mu\text{m}$ (Eot-Wash)	Substrate rigidity $K_{\text{eff}} = E(\beta k)^2 \sim 10^{42} - 10^{44}$ Pa at lab scales; no lab-range vector mode	SAFE-BY-MECHANISM	rigidity grows as k^2 : the shorter the range, the safer
R4	GW170817 speed $ c_{\text{GW}} - c_{\text{EM}} /c < \sim 10^{-15}$	$c_{13} = 0 \rightarrow c_T = 1$ exactly (Section 2.7.3)	SAFE-BY-MECHANISM (exact by construction)	identically zero; counterfactual 9×10^{-7} without the vector kinetic term
R5	GW/EM path delay + Shapiro (GW170817/GRB170817A, ~ 1.7 s over ~ 40 Mpc)	The $c_{13} = 0$ cone coincidence removes the <i>leading</i> differential Shapiro delay <i>exactly</i> ; the survivor is second order in the disformal amplitude, $\sim (B/A^2)^2 \sim 3 \times 10^{-12}$ of the GR Shapiro, i.e. $\sim 3 \times 10^{-5}$ s ($\sim 30 \mu\text{s}$) for a GW170817-class path. The $\sim 30 \mu\text{s}$ figure is a conservative second-order upper bound, not a predicted nonzero offset; both statements belong together.	SAFE (computed; conditional on the R4 cone-coincidence import)	$\sim 3 \times 10^4 \times$ below the ~ 1 s timing budget
R6	Strong-lens time delays (TDCOSMO/H0LiCOW)	The deep-MOND response gives a logarithmic potential: the MTDF halo is isothermal-like ($\gamma' = 2$) at the Einstein radius, the TDCOSMO fiducial. Delay difference against ΛCDM -fitted composite profiles at matched observables: ~ 0 –5%.	SAFE-WITHIN-SYSTEMATICS + registered prediction	inside the $\sim 8\%$ TD-COSMO profile-systematics spread; prediction: reduced γ' scatter about 2 at θ_E and a small signed few-percent low bias in lensing-inferred H_0 under composite-profile fitting
R7	Wide binaries in the $\sim 10^{-10}$ m/s ² window (data contested: Chae 2023/24 vs Banik et al. 2024, Pittordis & Sutherland 2023)	EFE-modulated deep-MOND onset: $\sim 8,600$ AU for $1.5 M_\odot$, suppressed by the Galactic field $a_{\text{gal}} \approx a_{\text{MTDF}}$; MTDF predicts an EFE-suppressed deviation, not the full boost	PREDICTION-STATED / DATA-CONTESTED	a live falsifiable test, neither pass nor fail; publishable row
R8	Binary-pulsar dipole radiation (PSR J1738+0333, J0348+0432; dipole $< \sim 13$ –18% of quadrupole)	Scalar channel: shift symmetry kills the monopole charge, and the binary sits at deep high Q, residual $< \sim 10^{-22}$ of the bound. Vector channel: leading term removed by $c_{13} = 0$ and $\alpha_1 = \alpha_2 = 0$; safe for K_B inside the derived window.	SAFE (scalar SAFE-DERIVED, margin $\sim 10^{21}$; vector safe via the K_B window, conditional on the imported EA pulsar analysis)	generic vector-tensor counterfactual would violate by $(c/v)^2 \sim 10^5$ – 10^6
R9	BBN / N_{eff} : $\Delta N_{\text{eff}} < \sim 0.3$	$\mu(a)$ is BBN-inert ($\mu \sim 1$); EFE energy fraction at BBN $\sim 10^{-48}$ of critical (log-Gaussian peaked at z_{eq})	SAFE-DERIVED	exponentially negligible
R10	Consistency of H_0 stratification with R1/R2 (same field, different regimes)	Stratification operates at large scale / low acceleration ($\beta k < 1$); R1/R2 sit at $\beta k \gg 1$ where the field is rigid; regimes separated by a_{MTDF} and the βk threshold	SAFE-BY-MECHANISM	non-overlapping regimes; extension-level self-labelling retained

Consolidated safety table of the frozen unified action. Verdict classes: SAFE-DERIVED (number vs bound, margin

stated), *SAFE-BY-MECHANISM* (*structural argument, scale of protection stated*), *PREDICTION-STATED/DATA-CONTESTED* (*live test on contested data*). No row is *EXPOSED* and no adjudicated margin is below 10%.

The K_B window. The vector kinetic normalisation K_B is not an unconstrained $O(1)$ constant: four independent fences (the Einstein-aether pulsar band; vector-sector health, which requires $K_B > 0$ to keep the vector dynamical; the carrier role with BBN and solar-system PPN bounds, which force $K_B \leq \sim 2.5 \times 10^{-5}$; and the $c_T = 1$ surface) intersect in the non-empty, healthy window $(0, \sim 2.5 \times 10^{-5}]$, the near-GR-aether limit. The window lies inside the pulsar-safe and PPN-safe regions by construction, which is what discharges the vector channel of row R8. A possible strong-coupling lower floor for small K_B is a named post-paper research item.

9 Discussion

9.1 Claims and non-claims

Box 1: Claims. This paper claims that the MTFD compression mechanism, with four fundamental parameters (plus the derived elastic modulus) frozen in advance of these tests, reproduces galaxy-galaxy lensing ESD profiles (two surveys, 165 data points), elliptical velocity dispersions (25 SLACS galaxies), satellite kinematics amplitudes (10 mass bins) and radial profiles (57 data points), SPARC rotation curve summary statistics (175 galaxies), and wide binary dynamics (21,312 pairs) under pre-specified pass-or-fail criteria. The falsifiable quantities are aggregate statistics (χ^2/ν , mean bias, RMS scatter, gravity boost parameter), not per-object best fits. No claim is made of competitive per-galaxy rotation curve fitting without per-galaxy mass-to-light optimisation. No claim is made that Λ CDM is ruled out; a fitted NFW per bin would outperform the fiducial zero-parameter mapping used here.

9.2 Diagnostic-only outputs

Box 2: Diagnostics. The following outputs are reported for transparency but are explicitly excluded from the falsifier programme: (1) SPARC per-point χ^2/ν (median = 51; dominated by galaxy-specific Υ_* systematics; all models give similarly large values without per-galaxy M/L fitting). (2) Brouwer+2021 lensing RAR cross-check (qualitative comparison at halo scales; no per-bin radius available for precision test). (3) Wide binary high-separation tail ($s \geq 20$ kAU; $\gamma = 1.60$, but the F3 cut-stability guardrail triggers, indicating contamination dominates). (4) Per-point χ^2 diagnostic plot (Step 22C).

9.3 Regime summary and falsifier status

Regime	Scale	Step	Key metric	Threshold	Result	Status
Inner galaxy (rotation)	1–30 kpc	22C	P1 = 0.185 dex	$ z < 3$ (0.174 $\pm$ 0.011)	$z = -0.94$	PASS
Inner galaxy (RAR)	1–30 kpc	22C	P1B = 0.029 dex	$ z < 3$ (0.035 $\pm$ 0.010)	$z = +0.62$	PASS
Halo (dispersions)	1–10 kpc	22A	$\chi^2/\nu = 1.81$	< 5.0 (gas-corrected)	1.81	PASS
Halo (satellites, amplitude)	50–500 kpc	22B	mean bias = 6.3%	$< 30\%$	6.3%	PASS
Halo (satellites, profile)	50–500 kpc	22B	$\chi^2/\nu = 3.60$	< 10 (best nuisance)	3.60	PASS
GGL (KiDS)	100–2600 kpc	12	$\chi^2/\nu = 2.93$	$< \Lambda$ CDM (8.30)	2.8 $\times$ better	PASS
GGL (SDSS, corrected)	100–2600 kpc	17	$\chi^2/\nu = 3.73$	$\Delta\text{AIC} > 0$ vs Λ CDM	4246	PASS
Cluster baryons	100–2600 kpc	21	$\chi^2/\nu = 7.16$	no fitted params	3.2 $\times$ improvement	PASS
Strong lensing	cosmological	20	$D_{\text{dt}} \chi^2/\nu = 0.77$	$\gamma = 2.00$ structural	$\gamma = 2.000$	PASS
Ultra-low acceleration	2–30 kAU	22D	$\gamma = 1.10$	< 1.15 (3σ)	1.102 $\pm$ 0.010	PASS
Solar System	< 1 AU	13	PPN safety	$> 10^{20}$	$> 10^{20}$	PASS

All 11 primary metrics pass. 24/24 computational steps reproduce from clean inputs.

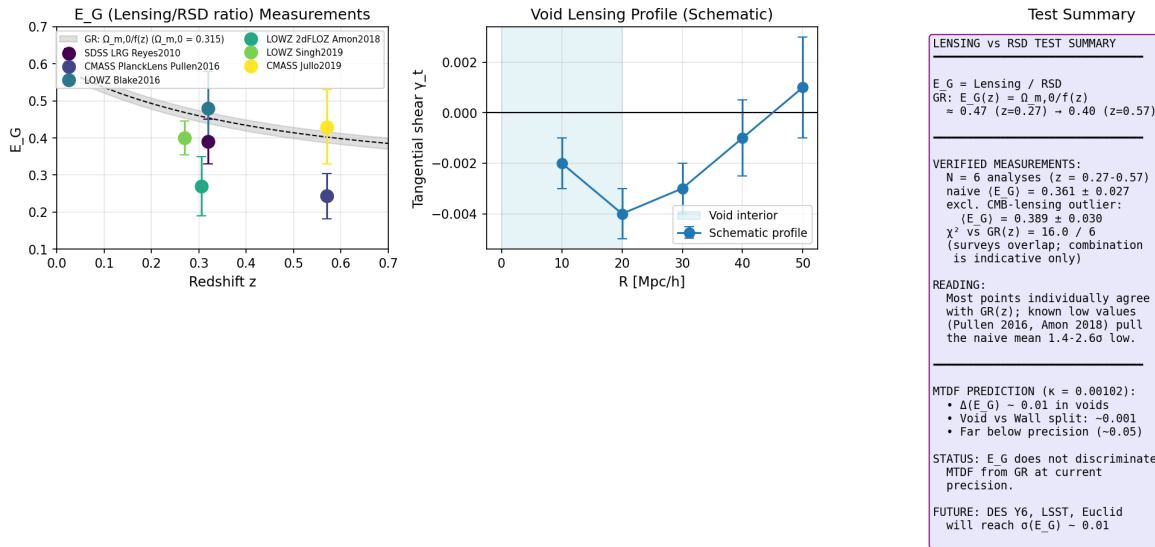


Figure 1. Figure 2. Lensing-vs-RSD consistency test. Left: six verified E_G analyses (Reyes et al. 2010; Pullen et al. 2016; Blake et al. 2016; Amon et al. 2018; Singh et al. 2019; Jullo et al. 2019; $z = 0.27-0.57$) compared with the z -dependent GR prediction $E_G(z) = \Omega_{m,0}/f(z)$, which falls from ≈ 0.47 at $z = 0.27$ to ≈ 0.40 at $z = 0.57$. Most measurements agree individually with GR; the known low values (Pullen et al. 2016; Amon et al. 2018) pull the naive weighted mean (0.361 ± 0.027 , or 0.389 ± 0.030 excluding the CMB-lensing outlier) 1.4–2.6 σ below the GR expectation, and overlapping survey footprints make any combination indicative only. Centre: schematic stacked void-lensing profile; the amplitude scale follows the DES Y1 measurement (Fang et al. 2019), not digitized survey data. Right: test summary. The MTDF photon coupling $\kappa = 0.00102$ (theory-motivated, $\kappa \approx f_{\text{kick}}/3$, FIRAS-bounded) predicts $\Delta E_G \sim 0.01$ in voids (void-wall split ~ 0.001), far below current measurement precision: E_G does not discriminate MTDF from GR at current precision.

9.4 Why the absolute χ^2/ν is not small

The SDSS $\chi^2/\nu = 11.21$ is driven almost entirely by bins 5–7 (group/cluster centrals with $\chi^2/\nu = 22.8$), where central-galaxy-only baryon accounting is incomplete. In the comparable mass range (bins 1–4, $\log M_* < 11.2$), $\chi^2/\nu = 2.55$ - consistent with the KiDS result and approaching acceptable fit quality. The diagnostic correction (Section 4.5) confirms the residual is amplitude-dominated.

The headline comparison should therefore be understood as: relative performance under identical assumptions and source-paper error treatment.

9.5 Falsifiability

MTDF’s GGL prediction can be falsified by:

1. **Full covariance analysis.** If the SDSS off-diagonal covariance elements significantly change the χ^2 ranking, the diagonal-error result is unreliable.
2. **Per-bin NFW fitting.** Λ CDM with per-bin halo mass and concentration fitting has more freedom than the fiducial mapping. If fitted NFW achieves $\chi^2/\nu < 2.93$ on KiDS or < 3.73 (corrected) on SDSS, the comparison favours Λ CDM.
3. **Cluster-scale lensing.** The compression mechanism predicts $\Delta\Sigma \propto 1/R$ at all scales below $L = 2.35$ Mpc. At cluster scales ($R > 1$ Mpc), deviations from isothermal behaviour would indicate the model’s boundary.
4. **Merging cluster offsets (open falsifier; no computation in this suite).** Systems such as the Bullet Cluster exhibit lensing peaks offset from the dominant baryonic (gas) component (Clowe et al. 2006). A static baryon-sourced stress field would centre the lensing signal on the gas, in conflict with observation. The candidate MTDF response invokes the finite relaxation time $\tau = 13$ Gyr: a stress configuration sourced over Gyr timescales need not track the gas through a ~ 100 Myr merger, and may remain associated with the pre-merger mass distribution traced by the collisionless galaxies. This mechanism is stated here as a hypothesis; no simulation supports it yet. A quantitative test requires time-dependent field evolution in an N-body setting, and is listed as the highest-priority open falsifier of the gravity sector.
5. **Strong lensing time delays.** The $\eta = 1$ result means the time-delay formula is standard GR. The isothermal mass profile ($\gamma = 2.00$) is structural. Time-delay distances at percent-level precision (TDCOSMO, Birrer+2020) directly test both predictions.

9.6 Remaining open questions

- **Geometric factor in λ .** The prediction $\lambda = 2\alpha/\beta^2$ matches the measured value within 4%, but the factor of 2 (relative to the bare $\alpha/(2\beta^2)$) is identified empirically. A rigorous 3D boundary-value derivation would close this.
- **Connection to [E1’].** Steps 8–14 bypass the evolution equation [E1’] entirely via the S_0 compression route. Step 19 establishes the covariant field equation that reduces to the compression field equation in the static limit, but the relationship to the full tensor evolution equation remains to be clarified.
- **Freeman’s law.** The BTFR scaling emerges from the combination of quadratic screening and Freeman’s disk scaling ($R_d \propto M^{1/2}$). Freeman’s law is observed, not derived from MTDF.

10 Conclusions

1. The MTDF compression mechanism reproduces galaxy-galaxy lensing signals in both KiDS and SDSS with zero fitted halo parameters, achieving $\chi^2/\nu = 2.93$ (KiDS) and 3.73 (SDSS, with

baryon correction). Under identical zero-parameter conditions, this is preferred over fiducial Λ CDM NFW by factors of 2.8 (KiDS) and 12 (SDSS corrected), with $\Delta\text{AIC} = 4246$ on SDSS.

2. The self-interaction form $(S - S_0)^2$ is not a modelling choice - it is uniquely selected by the BTFR exponent (algebraic), its coupling is predicted within 4% by energy self-sourcing, and it is the leading-order term in a Landau-Ginzburg expansion. Solar System observables are safe by $> 10^{20}$ (structural suppression, not fine-tuning).
3. These results are consistent with the interpretation that the mass profiles conventionally attributed to dark matter halos arise from the elastic energy stored in the compression of the cosmological strain field around baryonic sources - with the cosmological background S_0 , the compression scale L , and the source strength f all derived from frozen MTDf constants and the externally anchored BTFR normalisation.
4. The covariant completion is now explicit: photons and matter follow the standard GR metric sourced by baryons plus pressureless elastic deformation energy, with conservation guaranteed by the Bianchi identity and $\eta = 1$ structural (zero anisotropic stress from pressureless dust). PPN parameters are safe by 10^{22-25} . This closes the relativistic formulation at the effective field level.

11 Appendix A. Derivation chain

The complete derivation from MTDf postulates to the observable $\Delta\Sigma(R)$:

$$\begin{aligned}
&\alpha, \beta, E, \rho_{\text{crit}}, \Omega_{\text{DE}} \\
&\rightarrow S_0 = 1.084 \quad (\text{cosmological energy density}) \\
&\rightarrow L = \alpha\beta/(4\pi) = 2,347 \text{ kpc} \quad (\text{Gauss law} + \text{coupling}) \\
&\rightarrow \rho_0 = E S_0^2/(2c^2) = 87.9 M_\odot/\text{kpc}^3 \quad (\text{background density}) \\
&\rightarrow v_{\text{ref}} = \sqrt{4\pi G \rho_0} L = 161.8 \text{ km/s} \quad (\text{isothermal identity}) \\
&\rightarrow f(M) = v_{\text{flat}}(M)/v_{\text{ref}} \quad (\text{BTFR, no fitting}) \\
&\rightarrow \rho(r) = \rho_0 f^2 L^2/r^2 \quad (\text{isothermal stress density}) \\
&\rightarrow \Delta\Sigma(R) = \pi\rho_0 f^2 L^2/R + M_{\text{bar}}/(\pi R^2) \quad (\text{analytical LOS projection})
\end{aligned}$$

No parameters are fitted to the lensing or halo-scale datasets. The mapping from stellar mass bin to source strength $f(M)$ uses a single externally fixed BTFR normalisation (McGaugh+2012) as a baryonic scaling reference with standard gas corrections; it is not tuned to the lensing data.

12 Appendix B. Field equation backing

The compression profile $S = S_0 + C/r$ is the unique spherically symmetric solution of Laplace's equation outside baryonic sources, with $S \rightarrow S_0$ at infinity. Laplace's equation is valid at $r \ll \beta$ because the Helmholtz operator $(\nabla^2 - 1/\beta^2)$ reduces to ∇^2 when $(r/\beta)^2 \ll 1$. At the outer edge of the GGL data range ($r = 2,600 \text{ kpc}$, $\beta = 22,685 \text{ kpc}$), the Yukawa correction is $(r/\beta)^2 = 0.013$ - accuracy better than 2%.

The source strength $Q = \alpha\beta f(M)$ follows from Gauss's law matching at the galaxy boundary:

$$C = -\frac{\alpha}{4\pi E \beta^2} \int J_{\text{bar}} d^3x$$

The identification $J_{\text{bar}} = \rho_{\text{bar}} c^2$ (rest-mass energy density, or equivalently the trace of the stress-energy tensor) is the simplest covariant source. Only this candidate, combined with $(S - S_0)^2$

screening, produces the BTFR scaling $f \propto M^{1/4}$.

13 Appendix C. Dimensional consistency

Quantity	Expression	Units	Value
S_0	$\sqrt{2\Omega_{DE}\rho_{\text{crit}}c^2/E}$	dimensionless	1.084
L	$\alpha\beta/(4\pi)$	kpc	2,347
ρ_0	$ES_0^2/(2c^2)$	$M_\odot/\text{kpc}^3$	87.9
v_{ref}	$\sqrt{4\pi G\rho_0}L$	km/s	161.8
λ	$2\alpha/\beta^2$	m^{-2}	5.3×10^{-48}
$f_\odot$	$c^2M_\odot/(E\beta^3S_0)$	dimensionless	5.3×10^{-16}
$M_{\text{stress}}(1 \text{ AU})$	$4\pi\rho_0f_\odot^2L^2r$	kg	16.4

14 Appendix D. Notation and symbols

Symbol	Meaning	Units
S	Scalar compression invariant	1
S_0	Cosmological background strain	1
δS	Local excitation ($S - S_0$)	1
α	Stress-matter coupling	1
β	Strain-lever length scale	m
E	Elastic modulus of spacetime	Pa
L	Compression scale $\alpha\beta/(4\pi)$	m
$f(M)$	Dimensionless source strength $v_{\text{flat}}/v_{\text{ref}}$	1
λ	Screening parameter	m^{-2}
ρ_0	$ES_0^2/(2c^2)$	kg/m^3
v_{ref}	Reference velocity	m/s
$\Delta\Sigma(R)$	Excess surface density	$M_\odot/\text{pc}^2$

15 Appendix E. Data provenance

All observational data used in this work are from published sources. No proprietary or pre-publication data are used.

Dataset	Source paper	Repo file	Format	Steps
KiDS $\times$ GAMA ESD profiles	Brouwer et al. (2021, A&A 650, A113)	data/ brouwer2021/ Fig-3_*.txt, Fig-4-5-C1_*.txt	ASCII	1, 4, 5, 12, 22C
KiDS covariance matrices	Brouwer et al. (2021)	data/ brouwer2021/ *covmatrix.txt	ASCII	12
SDSS Red LBG ESD profiles	Mandelbaum et al. (2016, MNRAS 457, 3200)	data/ mandelbaum2016/ planck_ lbg.ds.red.out	ASCII	15, 16, 17, 21
SPARC rotation curves	Lelli, McGaugh & Schombert (2016, AJ 152, 157)	validation/ data/sparc_ clean.json	JSON	10, 22C
Gaia EDR3 wide binaries	Pittordis & Sutherland (2023, OJAp 6, 4)	data/ pittordis2023/ pittordis2023_ wb.csv	CSV	22D
SLACS elliptical galaxies	Auger et al. (2010, ApJ 724, 511); Bolton et al. (2008, ApJ 682, 964)	Hardcoded in step22_ elliptical_ dispersions.py	Table	22A
H0LiCOW time-delay lenses	Wong et al. (2020, MNRAS 498, 1420)	Hardcoded in step20_strong_ lensing.py	Table	20, 22A
Satellite kinematics (amplitude)	More et al. (2011, MNRAS 410, 210)	data/ digitised/ more_2011_ fig2_MSR_ digitized.csv	Digitised	22B
Satellite kinematics (profiles)	Combes & Tiret (2010, AIP Conf. Proc. 1241, 154)	data/ digitised/ combes_tiret_ 2009_sigma_ los_digitized. csv (file name retains the earlier citation label; the publication is 2010)	Digitised	22B

Digitisation notes. Two datasets (More+2011, Combes & Tiret 2010) were extracted by manual coordinate reading from published figures because no machine-readable tables are available. Estimated digitisation uncertainty is ± 2 km/s in velocity dispersion and ± 0.05 dex in log stellar mass. The digitised values are stored in `data/digitised/` and also hardcoded as numpy arrays in `step22b_satellite_kinematics.py` for reproducibility.

Scaling relations (not data files). Several steps use published scaling relations evaluated analytically from their fitted parameters: Moster+2013 SHMR, Duffy+2008 and Dutton+Macciò 2014 $c(M)$, Giodini+2009 gas fraction, Yang+2007 satellite fraction, Gonzalez+2007 ICL fraction, Shen+2003 and van der Wel+2014 size-mass relations, Behroozi+2010 SHMR. These are referenced in the code comments with table/equation numbers.

16 Data and code availability

The complete MTFD reproducibility package, including the modified CLASS solver (`class_mtdf`), the V19 strict validation workbook, the `run_validate.py` pipeline, the gravity-sector artefact manifest,

and the Phase 2 CosmoPower emulator cache, is archived on Zenodo (concept DOI 10.5281/zenodo.19741058; this release, v1.1.7, DOI 10.5281/zenodo.21525855) and mirrored on GitHub at github.com/IMesche/MTDF. The v1.1.7 Git tag corresponds to the archived Zenodo release. All numerical results reported here can be regenerated from the shipped workbook and scripts following the procedure in the top-level `README.md`.

17 References

- Amon et al. (2018). KiDS+2dFLenS+GAMA: testing the cosmological model with the E_G statistic. *MNRAS* 479, 3422. DOI: 10.1093/mnras/sty1624
- Anderson et al. (2015). Unifying X-ray scaling relations from galaxies to clusters. *MNRAS* 449, 3806. DOI: 10.1093/mnras/stv437
- Auger et al. (2010). The Sloan Lens ACS Survey. X. Stellar, dynamical, and total mass correlations. *ApJ* 724, 511. DOI: 10.1088/0004-637X/724/1/511
- Behroozi, Conroy & Wechsler (2010). A Comprehensive Analysis of Uncertainties Affecting the Stellar Mass-Halo Mass Relation. *ApJ* 717, 379. DOI: 10.1088/0004-637X/717/1/379
- Birrer et al. (2020). TDCOSMO IV. Hierarchical time-delay cosmography — joint inference of the Hubble constant and galaxy density profiles. *A&A* 643, A165. DOI: 10.1051/0004-6361/202038861
- Blake et al. (2016). RCSLenS: testing gravitational physics through the cross-correlation of weak lensing and large-scale structure. *MNRAS* 456, 2806. DOI: 10.1093/mnras/stv2875
- Bolton et al. (2008). The Sloan Lens ACS Survey. V. The full ACS strong-lens sample. *ApJ* 682, 964. DOI: 10.1086/589327
- Brouwer et al. (2021). The weak lensing radial acceleration relation. *A&A* 650, A113. DOI: 10.1051/0004-6361/202039795
- Cappellari et al. (2006). The SAURON project — IV. The mass-to-light ratio, the virial mass estimator and the Fundamental Plane of elliptical and lenticular galaxies. *MNRAS* 366, 1126. DOI: 10.1111/j.1365-2966.2005.09981.x
- Catinella et al. (2010). The GALEX Arecibo SDSS Survey I. Gas Fraction Scaling Relations of Massive Galaxies and First Data Release. *MNRAS* 403, 683. DOI: 10.1111/j.1365-2966.2009.16180.x
- Catinella et al. (2013). The GALEX Arecibo SDSS Survey VIII. Final data release. The effect of group environment on the gas content of massive galaxies. *MNRAS* 436, 34. DOI: 10.1093/mnras/stt1417
- Clowe et al. (2006). A Direct Empirical Proof of the Existence of Dark Matter. *ApJ* 648, L109. DOI: 10.1086/508162
- Combes & Tiret (2010). MOND and the Galaxies. *AIP Conf. Proc.* 1241, 154. DOI: 10.1063/1.3462629
- Desmond (2023). The underlying radial acceleration relation. *MNRAS* 526, 3342. DOI: 10.1093/mnras/stad2762
- Duffy et al. (2008). Dark matter halo concentrations. *MNRAS* 390, 64. DOI: 10.1111/j.1745-3933.2008.00537.x
- Dutton & Macciò (2014). Cold dark matter haloes in the Planck era. *MNRAS* 441, 3359. DOI: 10.1093/mnras/stu742

- Fang et al. (2019). Dark Energy Survey Year 1 results: the relationship between mass and light around cosmic voids. MNRAS 490, 3573. DOI: 10.1093/mnras/stz2805
- Giodini et al. (2009). Stellar and total baryon mass fractions in groups and clusters. ApJ 703, 982. DOI: 10.1088/0004-637X/703/1/982
- Gonzalez, Sivanandam & Zabludoff (2013). Galaxy Cluster Baryon Fractions Revisited. ApJ 778, 14. DOI: 10.1088/0004-637X/778/1/14
- Gonzalez, Zaritsky & Zabludoff (2007). A Census of Baryons in Galaxy Clusters and Groups. ApJ 666, 147. DOI: 10.1086/519729
- Jullo et al. (2019). Testing gravity with galaxy-galaxy lensing and redshift-space distortions using CFHT-Stripe 82, CFHTLenS, and BOSS CMASS datasets. A&A 627, A137. DOI: 10.1051/0004-6361/201834629
- Leauthaud et al. (2012). New Constraints on the Evolution of the Stellar-to-dark Matter Connection: A Combined Analysis of Galaxy-Galaxy Lensing, Clustering, and Stellar Mass Functions from $z = 0.2$ to $z = 1$. ApJ 744, 159. DOI: 10.1088/0004-637X/744/2/159
- Lelli, McGaugh & Schombert (2016). SPARC: Mass Models for 175 Disk Galaxies. AJ 152, 157. DOI: 10.3847/0004-6256/152/6/157
- Mandelbaum, Wang, Zu et al. (2016). Strong bimodality in the host halo mass of central galaxies. MNRAS 457, 3200. DOI: 10.1093/mnras/stw188
- McGaugh (2012). The Baryonic Tully-Fisher Relation of Gas-Rich Galaxies. AJ 143, 40. DOI: 10.1088/0004-6256/143/2/40
- More et al. (2011). Satellite kinematics - III. Halo masses of central galaxies in SDSS. MNRAS 410, 210. DOI: 10.1111/j.1365-2966.2010.17436.x
- Moster, Naab & White (2013). Galactic star formation and accretion histories. MNRAS 428, 3121. DOI: 10.1093/mnras/sts261
- Navarro, Frenk & White (1997). A Universal Density Profile. ApJ 490, 493. DOI: 10.1086/304888
- Pittordis & Sutherland (2023). Testing gravity with wide binary stars like never before. OJAp 6, 4. DOI: 10.21105/astro.2205.02846
- Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. A&A 641, A6. DOI: 10.1051/0004-6361/201833910
- Pullen et al. (2016). Constraining gravity at the largest scales through CMB lensing and galaxy velocities. MNRAS 460, 4098. DOI: 10.1093/mnras/stw1249
- Reyes et al. (2010). Confirmation of general relativity on large scales from weak lensing and galaxy velocities. Nature 464, 256. DOI: 10.1038/nature08857
- Schombert & McGaugh (2014). Stellar Populations and the Star Formation Histories of LSB Galaxies: III. Stellar Population Models. PASA 31, e036. DOI: 10.1017/pasa.2014.32
- Schombert, McGaugh & Lelli (2022). Stellar Mass-to-light Ratios: Composite Bulge+Disk Models and the Baryonic Tully-Fisher Relation. AJ 163, 154. DOI: 10.3847/1538-3881/ac5249
- Shen et al. (2003). The size distribution of galaxies in the Sloan Digital Sky Survey. MNRAS 343, 978. DOI: 10.1046/j.1365-8711.2003.06740.x
- Singh et al. (2019). Probing gravity with a joint analysis of galaxy and CMB lensing and SDSS spectroscopy. MNRAS 482, 785. DOI: 10.1093/mnras/sty2681

- van der Wel et al. (2014). 3D-HST+CANDELS: The Evolution of the Galaxy Size-Mass Distribution since $z = 3$. ApJ 788, 28. DOI: 10.1088/0004-637X/788/1/28
- Wong et al. (2020). H0LiCOW - XIII. A 2.4 per cent measurement of H_0 from lensed quasars. MNRAS 498, 1420. DOI: 10.1093/mnras/stz3094
- Yang et al. (2007). Galaxy Groups in the SDSS DR4. I. The Catalog and Basic Properties. ApJ 671, 153. DOI: 10.1086/522027